

Class 5: Cash Flow Analysis

PRACTICE QUESTIONS: SOLUTIONS

These practice questions are designed to help improve your understanding of the course material. The questions referenced in this document can be found at the **end of each chapter in your textbook**.

Use the **Table of Contents** below to identify the list of questions and the **suggested time** for completing each one. When using this resource:

- ☐ **Do the exercises soon after the lecture** and ideally **before tutorials**. Studies on learning show that practicing material shortly after it's introduced improves understanding and retention. Spacing out your practice and reviewing regularly also strengthens long-term memory.
- ☐ **Attempt the exercises first** without looking at the solutions or debriefing notes. This will help you simulate real exam conditions and identify areas where you may need further practice.
- ☐ Be **conscious of the suggested time** for each question. Staying aware of time management is a valuable skill for exams.
- ☐ After completing the exercises, use the solutions provided to **debrief your work**. Compare your answers, understand any mistakes, and learn from them to solidify your understanding of the concepts.

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CH. 11 | REPORTING AND ANALYZING CASH FLOWS

MINI EXERCISES (~ 65 min)

M11-22 (~ 5 min)

- a. Cash flow from an operating activity.
- b. Cash flow from an investing activity.
- c. Cash flow from an investing activity.
- d. Cash flow from an operating activity.
- e. Cash flow from a financing activity.
- f. Cash flow from a financing activity.
- g. Cash flow from an investing activity.

M11-24 (~ 5 min)

- a. (3) Cash flow from a financing activity.
- b. (1) Cash flow from an operating activity.
- c. (4) Noncash investing and financing activity.
- d. (1) Cash flow from an operating activity.
- e. (1) Cash flow from an operating activity.
- f. (5) None of the above (a change in the composition of cash and cash equivalents).

M11-25 (~ 20 min)

TEMPLATE TO USE:

a. Beyer GMBH

	Balance Sheet									Income Statement						
Transaction	Cash Asset	+	Accts. Receiv-able	+	Inven-tories	=	Accts. Payable	+	Contrib. Capital	+	Earned Capital	Revenue	-	Expenses	=	Net Income

a. Beyer GMBH

Transaction	Balance Sheet									Income Statement						
	Cash Asset	+	Accts. Receivable	+	Inventories	=	Accts. Payable	+	Contrib. Capital	+	Earned Capital	Revenue	-	Expenses	=	Net Income
(1)		+	+507,400	+		=		+		+	+507,400	+507,400	-		=	+507,400
(2)	+91,500	+		+		=		+		+	+91,500	+91,500	-		=	+91,500
(3)		+		+	-320,100	=		+		+	-320,100		-	+320,100	=	-320,100
(4)		+		+	-63,400	=		+		+	-63,400		-	+63,400	=	-63,400
(5)		+		+	+351,600	=	+351,600	+		+			-		=	
(6)	-47,700	+		+	+47,700	=		+		+			-		=	
(7)	+483,400	+	-483,400	+		=		+		+			-		=	
(8)	-340,200	+		+		=	-340,200	+		+			-		=	
(9)	-172,300	+		+		=		+		+	-172,300		-	+172,300	=	-172,300
Total	+14,700	+	+24,000	+	+15,800	=	+11,400	+		+	+43,100	+598,900	-	+555,800	=	+43,100

b. Net income was €43,100 (from the net income column), and cash flow from operating activities was €14,700 (from the cash column).

- c. 1. Accounts receivable increased by €24,000,
 2. Inventories increased by €15,800, and
 3. Accounts payable increased by €11,400.

d. The accounting equation is kept with every entry, so it is kept for the totals over the period.

$$\text{Cash flow} + \text{change in accounts receivable} + \text{change in inventory} \\ = \text{Change in accounts payable} + \text{net income}.$$

This relationship can be presented in the following indirect method cash flow from operating activities.

Net income	€ 43,100
- Change in accounts receivable	-24,000
- Change in inventories	-15,800
+ Change in accounts payable	+11,400
Cash flow from operating activities	€ 14,700

M11-28. (~ 15 min)

Net loss.....		\$(21,000)
Add (deduct) items to convert net loss to cash basis		
Add back depreciation.....	8,600	
Subtract change in operating assets:		
Accounts receivable.....	9,000	
Inventory.....	3,000	
Prepaid expenses.....	3,000	
Add change in operating liabilities:		
Accounts payable.....	4,000	
Accrued liabilities.....	(2,600)	
Net cash provided by operating activities.....	<u>\$ 4,000*</u>	

* If data are provided separately, the company will report the amount of interest paid and the amount of tax paid in this section.

Weber Company's operating activities provided \$4,000 cash.

The dividend paid to shareholders affects cash flows from financing activities.

M11-29 (~ 10 min)

A “+” indicates that the amount is added and a “-” indicates that it is subtracted when preparing the cash flow statement using the indirect method.

CHINA MOBILE LIMITED			
Consolidated Statement of Cash Flows – Selected Items			
1	Increase in accounts receivable	Operating	-
2	Capital expenditure	Investing	-
3	Proceeds from disposal of a joint venture	Investing	+
4	Gain on disposal of property, plant and equipment	Operating	-
5	Repayment of interest bearing borrowings	Financing	-
6	Increase in inventories	Operating	-
7	Increase in prepayments and other current assets	Operating	-
8	Proceeds from issuance of shares under share option scheme	Financing	+
9	Increase in accounts payable	Operating	+
10	Profit before taxation (note: this is like the Net Income line)	Operating	+
11	Acquisition of other intangible assets	Investing	-
12	Increase in accrued expenses and other payables	Operating	+
13	Dividends paid to the Company's equity shareholders	Financing	-
14	Depreciation of property, plant and equipment	Operating	+

M11-30 (a, b) (~10 min)

a.	Rent expense	\$ 60,000
–	Prepaid rent decrease	<u>(2,000)</u>
=	Cash paid for rent	<u>\$ 58,000</u>

Transaction	Balance Sheet						Income Statement			
	Cash	+	Noncash Assets	=	Liabilities	+	Contr. Capital	+	Earned Capital	
Begin Balance		+	10,000 Prepaid rent	=		+		+		- =
Make rent payment	-X	+	+X Prepaid Rent	=		+		+		- =
Record rent expense		+	-60,000 Prepaid Rent	=		+		+	-60,000 Retained Earnings	60,000 Rent Expense = -60,000
End Balance		+	8,000	=		+		+		- =

Key Concepts

- ☐ Cash Paid for Rent: The actual amount of cash the company has paid for rent during the accounting period. This is what we're trying to calculate.
- ☐ Prepaid Rent: An asset account representing rent payments made in advance for future periods. It appears on the balance sheet under current assets.
- ☐ Rent Expense: The cost of using rented facilities or equipment during the accounting period. This amount is recorded on the income statement.

Understanding the Relationship

The cash paid for rent can be different from the rent expense recognized in the income statement due to changes in prepaid rent. When you pay cash now, you increase prepaid rent. Conversely, as time passes and you use the space, the remaining benefit decreases and this is reflected in decreased prepaid rent. As a result, equity decreases, and we recognize rent expense.

Formula to Calculate Cash Paid for Rent

To find the cash paid for rent during the period, you can use the following formula:

- ☐ Beginning prepaid rent + Cash paid for rent – Prepaid rent “used” (expensed) = Ending prepaid rent
- ☐ Therefore, Cash paid for rent = Ending prepaid rent – Beginning prepaid rent + Rent expense
- ☐ Use the formula:
- ☐ Cash paid for rent = \$8,000 - \$10,000 + \$60,000 = \$58,000

b.	Interest income	\$ 16,000
–	Interest receivable increase	(700)
=	Cash received as interest	<u>\$ 15,300</u>

Transaction	Balance Sheet						Income Statement			
	Cash	+	Noncash Assets	=	Liabilities	+	Contr. Capital	+	Earned Capital	
Begin Balance		+	3,000 Interest receivable	=		+		+		- =
Record interest income		+	+16,000 Interest receivable	=		+		+	+16,000 Retained Earnings	+16,000 Interest income = +16,000
Receive interest payment	+X	+	-X	=		+		+		- =
End Balance		+	3,700	=		+		+		- =

Key Concepts

- ☐ Interest Receivable: An asset account representing interest that has been earned but not yet received in cash. It appears on the balance sheet under current assets. As we earn interest, interest receivable increases and we recognize the increase in equity as interest income. Remember, this occurs as interest is earned. So, interest receivable increases when interest is earned but not yet collected and decreases when cash is received for interest.
- ☐ Cash Received from Interest: The actual amount of cash the company has received from interest during the accounting period.
- ☐ Interest Income: The amount of interest earned by the company during the accounting period. This amount is recorded on the income statement and increases net income.

Understanding the Relationship

The relationship between **cash received from interest**, **interest income**, and **interest receivable** is as follows:

- ☐ **When interest income is earned, but cash has not been received**, the company records an **increase in Interest Receivable**.
- ☐ **When cash is received for interest**, the company records a **decrease in Interest Receivable** (if it was previously recorded as receivable) or recognizes cash for interest income earned and received in the same period.
- ☐ **Cash Received from Interest** may differ from the **Interest Income** recognized due to changes in the **Interest Receivable** balance.

Formula to Calculate Cash Received from Interest

To find the **Cash Received from Interest** during the period, you can use the following formula:

- ☐ Beginning interest receivable + Interest earned (i.e. Interest Income) – Cash received for interest = Ending interest receivable
- ☐ Therefore, Cash received for interest = Beginning interest receivable + Interest earned – Ending interest receivable
- ☐ Use the formula:
- ☐ Cash received for interest = \$3,000 + \$16,000 – \$3,700 = \$15,300

Check Your Understanding

Why is Cash Received from Interest less than Interest Income?

- ☐ Because some of the interest income earned has not yet been received in cash, increasing Interest Receivable

What does an increase in Interest Receivable indicate?

- ☐ That additional interest income was earned but not collected during the period.
- ☐ An **increase** means that more interest income was earned than cash received; thus, cash received is **less** than interest income.

EXERCISES (~ 40 min)

E11-35 (a only) (~ 10 min)

a. Net income	\$113,000
Add (deduct) items to convert net income to cash basis	
Accounts receivable increase	(5,000)
Inventory decrease	6,000
Prepaid insurance increase	(1,000)
Accounts payable increase	4,000
Wages payable decrease	(2,000)
Net cash provided by operating activities	<u>\$115,000*</u>

* If data are provided separately, the company will report the amount of interest paid and the amount of tax paid in this section.

E11-42 (~30 min)

RONALDO CORPORATION Statement of Cash Flows Year ended December 31, 2020		
Cash Flows from Operations:*		
Net income	\$ 700	
Adjustments:		
Add back Depreciation	350	
– Change in Accounts Receivable	(900)	
– Change in Inventory	(100)	
– Change in Prepaid Expenses	250	
+ Change in Accounts Payable	400	
+ Change in Income Taxes Payable	<u>(100)</u>	
Cash Flows from Operating Activities		\$ 600
Cash Flows from Investing:		
Purchases of Equipment	(1,200)	
Proceeds from Disposal of Equipment	<u>600</u>	
Cash Flows from Investing Activities		(600)
Cash Flows from Financing:		
Dividends Paid	(250)	
Increase in Short-term Debt	1,500	
Decrease in Long-term Debt	<u>(1,000)</u>	
Cash Flows from Financing Activities		<u>250</u>
Net Change in Cash		250
Beginning Cash Balance		300
Ending Cash Balance		<u>\$ 550</u>

* If data are provided separately, the company will report the amount of interest paid and the amount of tax paid in this section.

PROBLEMS (~ 15 min)**P11-54. (~ 15 min)****AstraZeneca Plc (USD in millions)**

- a. Depreciation, amortisation, and impairment are added back to undo the effect it had on the income statement. AstraZeneca deducted \$3,762 million of depreciation, amortisation, and impairment expense in computing net income. Depreciation, amortisation, and impairment are noncash expenses, so AstraZeneca did not actually use cash to pay those expenses in 2019. Hence, net income is lower than cash from operations by the depreciation, amortisation, and impairment amount. This add-back is NOT a source of cash as some mistakenly believe. Cash is generated by profitable operations, not by depreciation.
- b. Credit sales increase net income and trade receivables, but have no effect on cash flow. An increase in trade receivables reflects the fact that net income is higher than cash flow from operations. As a result, we subtract the \$898 increase in trade receivables from net income to compute cash from operations. It does not mean that the company used up \$898 for trade receivables.

An increase in inventories means that inventory purchases have exceeded goods sold for the period. Cash was used for these purchases (i.e., holding accounts payable constant) and, thus, net income is higher than cash flows from operations. Hence, we subtract the \$316 increase in inventories to net income to compute cash from operations. It does not mean that the company used \$316 for its inventory.

- c. The increase in trade and other payables and provisions reflects credit purchases in current period were more than payments for goods purchased in prior periods. All new purchases were sold (i.e., holding inventory constant) and, thus, net income was lower than cash flow (COGS was higher than cash paid). Hence, we add back the \$868 increase in trade payables to net income to compute cash from operations. It does not mean that the company received \$868 for trade payables.
- d. Capital expenditures (CAPEX) are necessary for routine equipment replacements and new additions, while depreciation expense represents the wear and tear of depreciable assets. In general, we should expect net CAPEX to exceed depreciation expense. In 2019, \$942 (=979 – 37) million was spent on new property, plant, and equipment, but depreciation was likely higher than that (Assume that depreciation is one-third of the total depreciation, amortisation and impairment of \$3,762 million). Hence, net CAPEX was not enough to cover the wear and tear on the property, plant, and equipment for the year.
- e. In 2019, AstraZeneca paid \$3,592 million in cash dividends (did not repurchase any shares). This may indicate that the company is in a mature stage, with strong cash inflows but limited growth opportunities. Hence, it returns excess cash to investors.

continued next page

P11-54. concluded

- f. AstraZeneca generated cash flows of \$2,969 million from operating activities. The company had a net cash outflow from investing activities of \$657 million, which was due primarily to purchases of intangible assets (\$1,481 million) and disposal of intangible assets (\$2,076). The company also had a net cash outflow from financing activities of \$1,765 million, mainly from its dividend payments and proceeds from share issuance. Overall, net cash increased by \$547 million. Hence, the company was generating enough cash flows from its operations to fund its investing and financing needs.